

COUNTEREXAMPLES TO GUI'S SORTING CONJECTURE FOR REDUCED KRONECKER COEFFICIENTS

ABSTRACT. Gui conjectured that splitting the decreasing rearrangement of the combined parts of two partitions into its odd- and even-indexed subsequences produces a coefficientwise larger stable tensor product. We disprove this conjecture with the infinite family

$$\bar{g}_{(k,1),(k-1)}^{(1)} = 0 < 1 = \bar{g}_{(k),(k-1,1)}^{(1)} \quad (k \geq 2).$$

The case $k = 2$ is, up to interchanging the factors, the unique counterexample among pairs of nonempty partitions with at most four boxes in total. The same family also disproves the multipartition extension stated as Conjecture 5.7 in the published version of Gui's paper.

Let $g_{\lambda,\mu}^\nu$ denote the ordinary Kronecker coefficient. For a partition α and all sufficiently large N , set

$$\alpha[N] = (N - |\alpha|, \alpha).$$

Murnaghan's stability theorem defines the reduced Kronecker coefficient by

$$\bar{g}_{\alpha,\beta}^\gamma = \lim_{N \rightarrow \infty} g_{\alpha[N],\beta[N]}^{\gamma[N]};$$

see [2].

For partitions λ and μ , let $(\delta_1, \delta_2, \dots)$ be the decreasing rearrangement of all their parts, and define

$$\text{sort}_1(\lambda, \mu) = (\delta_1, \delta_3, \dots), \quad \text{sort}_2(\lambda, \mu) = (\delta_2, \delta_4, \dots).$$

Gui's Conjecture 5.5 [1] asserts that

$$(1) \quad \bar{g}_{\text{sort}_1(\lambda,\mu),\text{sort}_2(\lambda,\mu)}^\nu \geq \bar{g}_{\lambda,\mu}^\nu \quad \text{for every partition } \nu.$$

For a partition α , let $R(\alpha)$ be the set of partitions obtained by deleting one removable corner. For a statement P , write $\mathbf{1}[P] = 1$ if P holds and $\mathbf{1}[P] = 0$ otherwise.

Lemma 1. *For all partitions α and β ,*

$$\bar{g}_{\alpha,\beta}^{(1)} = |R(\alpha) \cap R(\beta)| + \mathbf{1}[\alpha \in R(\beta)] + \mathbf{1}[\beta \in R(\alpha)].$$

Proof. For a partition $\sigma \vdash N$, write $[\sigma]$ for the corresponding irreducible S_N -module. The permutation module on N letters is $[N] \oplus [N-1, 1]$, and the tensor identity gives

$$[\sigma] \otimes \mathbb{C}^N \cong \text{Ind}_{S_{N-1}}^{S_N} \text{Res}_{S_{N-1}}^{S_N} [\sigma].$$

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The branching rule therefore yields, for $N \geq 2$ and $\sigma, \rho \vdash N$,

$$(2) \quad g_{\sigma, (N-1, 1)}^\rho = |R(\sigma) \cap R(\rho)| - \delta_{\sigma, \rho}.$$

For all sufficiently large N ,

$$R(\alpha[N]) = \{\alpha[N-1]\} \cup \{\tau[N-1] : \tau \in R(\alpha)\},$$

and likewise for β . By symmetry of ordinary Kronecker coefficients, (2) with $\sigma = \alpha[N]$ and $\rho = \beta[N]$ computes $g_{\alpha[N], \beta[N]}^{(N-1, 1)}$. The two first-row removals coincide exactly when $\alpha = \beta$, and this contribution is cancelled by the Kronecker delta. The lower-row removals contribute $|R(\alpha) \cap R(\beta)|$, while the two cross terms occur exactly when $\alpha \in R(\beta)$ or $\beta \in R(\alpha)$. Taking the stable value proves the formula. $\square$

Theorem 2. *For every integer $k \geq 2$,*

$$\bar{g}_{(k, 1), (k-1)}^{(1)} = 0, \quad \bar{g}_{(k), (k-1, 1)}^{(1)} = 1.$$

Consequently, Gui's Conjecture 5.5 is false.

Proof. The partitions (k) and $(k-1, 1)$ have the common one-box predecessor $(k-1)$ and no other common predecessor. Hence Lemma 1 gives

$$\bar{g}_{(k), (k-1, 1)}^{(1)} = 1.$$

The sizes of $(k, 1)$ and $(k-1)$ differ by two, so every term on the right-hand side of Lemma 1 vanishes. Thus

$$\bar{g}_{(k, 1), (k-1)}^{(1)} = 0.$$

Finally, the decreasing rearrangement of the parts of (k) and $(k-1, 1)$ is $(k, k-1, 1)$, and therefore

$$\text{sort}_1((k), (k-1, 1)) = (k, 1), \quad \text{sort}_2((k), (k-1, 1)) = (k-1).$$

At $\nu = (1)$, inequality (1) would therefore require $0 \geq 1$. $\square$

Corollary 3. *Up to interchanging the factors, $((2), (1, 1))$ is the unique pair of nonempty partitions (λ, μ) with $|\lambda| + |\mu| \leq 4$ for which (1) fails for some partition ν .*

Proof. For total size at most three, sorting fixes the pair up to interchange, so (1) holds with equality. At total size four, the only unordered pairs not fixed by sorting are

$$((1), (1, 1, 1)) \quad \text{and} \quad ((2), (1, 1)).$$

Gui's Theorem 5.6 [1] proves (1) for the first pair, since both factors are columns. The second pair fails by Theorem 2 with $k = 2$. $\square$

Corollary 4. *The multipartition statement published as Gui's Conjecture 5.7 is false.*

Proof. For two input partitions, the two partitions denoted $(\lambda \cup \mu)^{[1, 2]}$ and $(\lambda \cup \mu)^{[2, 2]}$ in [1, Conjecture 5.7] are precisely $\text{sort}_1(\lambda, \mu)$ and $\text{sort}_2(\lambda, \mu)$. Thus the two-factor case of that statement is Conjecture 5.5. $\square$

REFERENCES

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